

■ Music Career Simulator

Development Reports

Internal changelog & feature specification · April 2025

v1.1

Health & Fatigue Fixes

9 April · 11:08 PM

Song Quality Behaviour

- Quality depletion begins only after fatigue exceeds **75%** — quality never reaches 0 until fatigue hits 100%.
- Once fatigue crosses 100%, quality slowly decays to 0 over time rather than dropping instantly.

Starting Values

- Chosen genre starts at **60**; all other genres default to **30**.
- Chosen skill starts at **60**; all other skills default to **25**.

Song Creation Quality Range

- With all factors (fatigue, health, skills, genre) producing a base score x , the range is always $(x - 2.5, x)$.
- While $\text{max quality} \leq 5$: score is drawn uniformly at random within that range.
- Once max quality exceeds 5, probability distribution becomes:
 - 20% chance $\rightarrow (x - 2.5, x - 2)$ — shitty track territory
 - 60% chance $\rightarrow (x - 2, x - 1)$ — average output
 - 10% chance $\rightarrow (x - 1, x - 0.5)$ — solid track
 - 10% chance $\rightarrow (x - 0.5, x)$ — peak quality

→ Ensures realism: even skilled artists occasionally produce below-average work.

Health Depletion

- Health depletion has been **reduced** — previous rate was too aggressive.

Fatigue & Skill Progression

- Skill additions increase fatigue cost as skill value rises. Threshold to trigger a skill gain in the **90–100** range requires the artist to be below **20% fatigue**.
- Extrapolated fatigue requirements across all skill ranges (proportional scaling from the 90–100 anchor):
 - 0–30 range $\rightarrow$ no fatigue restriction
 - 30–50 range $\rightarrow$ below 70% fatigue
 - 50–70 range $\rightarrow$ below 55% fatigue
 - 70–80 range $\rightarrow$ below 40% fatigue
 - 80–90 range $\rightarrow$ below 30% fatigue
 - 90–100 range $\rightarrow$ below 20% fatigue

Fatigue Recovery & Other Adjustments

- Fatigue recovery speed increased — artists should be able to bounce back more easily.
- Active course enrolment adds an extra **+15% fatigue** per week for the full **12-week** duration.
- Song creation now causes **less fatigue** than before.

Live Module — Sentence Pool Expansion

- All live-module sentence pools have been expanded with additional varied sentences.
- Long sentences are no longer repeated cyclically.
- Comments per cycle increased from ~10 to **15–20**.
- Repeated live sessions with low-viewer counts will reflect reduced comment diversity accordingly.

v1.2

Money & Popularity System

10 April · 4:22 PM

Popularity Meter

- Introduced a **Popularity** meter that starts at a minimal value and grows as the artist releases songs and goes live.
- New modules: **Management** and **PR**.
- Going live grants a **+5% base popularity** boost per session.

Currency & Side Hustle Jobs

- Currency is denominated in **USD (\$)**.
- Side hustle jobs introduced — examples below (names changed to avoid direct branding):
 - Fast-food crew member — **\$20 / week**, low fatigue cost
 - Additional jobs scale: more pay = more fatigue per week
- Genre courses now have an **in-game monetary cost** attached.

Media Management & PR Contracts

- Artists can sign contracts with one of **5 management companies**. Contract durations: monthly, 6-month, or yearly.
- Base price is negotiable around each company's listed rate.

Company	Contract Cost	Popularity Boost	Tier
Amplify Collective	\$1,000 / year	10 – 20%	Entry
The Narrative Arc	\$3,000 / year	20 – 25%	Standard
Momentum Media	\$10,000 / year	40%	Professional
Catalyst Comms	\$40,000 / year	60%	Elite
Verve Public Relations	\$100,000 / year	75%	Prestige

→ *Management provides a weekly base popularity floor. Going live stacks an additional +5% on top of this.*

Song Releases & Popularity

- Releasing a song provides an instant **+5% base popularity** boost in that week.
- In the following week, the released song contributes popularity proportional to its quality rating (e.g. a 10/10 adds 10%, a 5/10 adds 5%).
- Releasing an **album** grants a one-time **+15% popularity** boost in the release week only.

Stream Count & Song Quality

- Streams are driven by both **popularity (70%)** and **song quality (30%)**.
- A low-quality song on high popularity will still underperform; a high-quality song on low popularity will still do respectably.
- New module: **Shawtify Streams** — displays stream counts for all released songs.

Stream Earnings Formula

- Base rate: **\$0.01 per stream**. Milestone bonuses stack as follows:

Milestone	Bonus Earnings	Running Total
Base (per stream)	\$0.01 / stream	—
Every 10k streams	+\$10	Cumulative
Every 100k streams	+\$500	Cumulative
1M streams	+\$3,000	~\$19,000 total

→ Example: 1M streams = $(1M \times \$0.01) + \$3,000 + (100 \times \$10) + (10 \times \$500) = \$19,000$

- If signed to management, the management company deducts a negotiated **revenue-share** % from weekly stream earnings.
- Weekly earnings summary is displayed after each simulation turn, including the management cut breakdown.

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Job & Popularity Fixes

10 April · 5:27 PM

Side Hustle Job Rework

- Jobs now lock the artist into a **12-week commitment** — you cannot switch or stack jobs during an active contract.
- Pay is disbursed **once per week**; the weekly-choice-and-multi-activation exploit has been removed.
- Fatigue cost per job has been increased across the board to better reflect the workload.

Management Base Popularity — Corrected Behaviour

- Base popularity from a management contract is now a **fixed weekly value** — it does not stack or accumulate over time.
- Example: if a company provides 20% popularity, the artist sits at 20% from that source every week of the contract.
- When the contract expires, base popularity **gradually fades** to 0 over several weeks rather than cutting off instantly.
- Weekly base popularity varies within the company's stated range for realism (e.g. 17.5% one week, 21% the next).

Song-Driven Weekly Popularity Gains

- Each song's stream count this week contributes popularity according to the table below. Each tier has an individual cap, and total song-popularity gain is capped at **20% per week**.

Stream Threshold	Popularity Gain	Weekly Cap
1,000+	+0.5%	2%
10,000+	+1%	4%

100,000+	+3%	8%
1,000,000+	+5%	10%
Total song popularity cap	20% / week	—

Worked Example

Song	Streams This Week	Popularity Gained
Song 1	3,000	+0.5%
Song 2	15,000	+1.0%
Song 3	130,000	+3.0%
Song 4	7,000,000	+5.0%
Song 5	3,000,000	+5.0%
Song 6	14,000,000	+0.0% (cap reached)
Total	—	14.5% (below 20% cap)

→ Total song popularity gain = 14.5%. Had it exceeded 20%, it would be capped at 20%.

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Fatigue, Recovery, Management & Song Creation

11 April · 3:16 PM

Fatigue Recovery — Revised Formula

- Base recovery is now a flat **50 points per week**: if last week ended at 0% fatigue, next week starts at 50%.
- Recovery when the artist was NOT at 0% fatigue last week is penalised by any recurring fatigue cost:
→ **Formula: Recovery = Base Recovery – (x% × Base Recovery) = 50 – (x × 50)**
- where x is the fatigue cost ratio of the active recurring action (e.g. a job costing 25% fatigue → x = 0.25).

Scenario	Last Week Fatigue	Job Cost	Next Week Starts At
Case 1 — fresh start	0%	24%	0% + 24% deducted directly = 24%
Case 2 — ongoing fatigue	100%	25%	$100 - (50 - 0.25 \times 50) = 100 - 37.5 = 62.5\%$
Case 2b — moderate fatigue	52%	24%	$52 - (50 - 0.24 \times 50) = 52 - 38 = 14\%$

→ This prevents the exploit where a rested player's fatigue immediately jumps up from recovery then gets slashed by the job cost every cycle.

Management Contract Pricing & Plan Rework

- All management company fees have been **substantially increased**; base popularity values have been slightly reduced.

- Three plan tiers now exist with different billing cadences:
 - **4-Week Plan** — billed every week (weekly charge applies each of the 4 weeks)
 - **6-Month Plan** — billed every 4 weeks (pay the 4-week rate once per 4-week period)
 - **Yearly Plan** — billed in two instalments: 6-month cost upfront, then again at week 24

Negotiation System — Conversational Chat Model

- The negotiation UI is now a **full back-and-forth conversation chat** rather than a single prompt.
- The company presents their initial ask (fee + revenue cut) at the top of the chat.
- The artist can propose a counter-offer adjusting **fee and/or revenue-cut %**; the company replies in character.
- Company reaction tiers based on the counter-offer:
 - Significantly below ask → **furious response** — company may walk away
 - Slightly below ask → mild pushback, possible compromise counter
 - At or near ask → standard acceptance
 - Above ask → **grateful public response** + internal monologue calling the artist a fool

Genre Course Costs — Updated Pricing

Proficiency Range	Course Cost
10 – 40	\$250
40 – 60	\$1,000
60 – 80	\$5,000
80 – 90	\$20,000
90 – 100	\$100,000

Song & Album Management — New Permissions

- **Delete songs** — songs kept as singles can now be deleted.
- **Album-released songs on Instagram Live** — songs released as part of an album now also appear in the released-songs section during live streams.
- **Add singles to an existing album** — songs previously kept as singles can be added to an already-created album.
- **Add released songs to an album** — fully released songs can be retroactively added to an album.

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Fine-Tuning Popularity & Fatigue

12 April · 4:14 PM

Going Live — Popularity Rework

- Each live session now grants a flat **+2 popularity**.
- Weekly cap from going live: **+6 popularity** (i.e. max 3 lives count per week).
- Live-sourced popularity **decays the following week** to 1–2 points rather than carrying over in full.

Song-Driven Popularity — Weekly Reset Model

- Song popularity contributions are now **recalculated fresh every week** based on that week's stream counts only — nothing carries over.

- Within a tier, the gain is flat regardless of exact stream count (e.g. a song doing 1k and one doing 9k both add **+0.5** — it crossed the 1k threshold, not the 10k one).
- Individual tier caps and the overall 20% weekly cap remain in effect.

Worked Example — Week-by-Week Song Popularity

Week	Song Streams	Popularity Gained	Carries to Next Week?
Week 7	Song A: 3,000 Song B: 7,000	0.5 + 0.5 = +1.0	No — resets to 0
Week 8	Song A: 1,000 Song B: 700	0.5 + 0.0 = +0.5	No — resets to 0

→ Song B hitting 700 in week 8 contributes nothing — it didn't cross the 1k threshold that week.

Job Fatigue Cases — Revised Logic & Pay Penalty

- Let x = the fatigue cost % of the job. Three cases now govern how the week starts and how much the artist is paid:

Case	Last Week Fatigue	Next Week Starts At	Pay Received
Case 1	0% (fully rested)	$x\%$ fatigue applied directly	Full pay (m)
Case 2	Exactly $x\%$ (job cost only)	Starts at $x\%$ again	Full pay (m)
Case 3	More than $x\%$ (extra fatigue)	$50 - (x \times 50)$ recovery applied eg. 50% fatigue + 20% job → starts at 10%	$m - (y/2)\% \times m$ where y = extra fatigue % above x

Pay Penalty Formula (Case 3)

→ $\text{Pay} = m - (y / 2)\% \times m$ where m = full pay, y = extra fatigue % above job cost

Job	Last Week Fatigue	Extra Fatigue (y)	Calculation	Pay Out
\$100 job, 20% cost	50%	30% above 20%	$100 - (0.30/2) \times 100 = 100 - 15$	\$85
\$100 job, 20% cost	20% (= x)	0%	No penalty	\$100

→ Player-facing message (Case 3): "You were extra tired last week — your boss docked some of your pay."

→ Sets up future mechanics: repeated Case 3 occurrences could eventually trigger a lay-off.

■ Happy 9 Months, Teena ji & Me! ■

12th April 2025 · Nine months of us · logged forever in a music sim dev doc, as it should be



Release Popularity Bump — Quality-Tiered, Week 1 Only

- The release bump now applies **only in the first week** of a song's release and is scaled to song quality.

Song Quality Range	Popularity Bump	Duration
0.0 – 3.0	+1	Week 1 only
3.1 – 5.0	+2	Week 1 only
5.1 – 7.0	+3	Week 1 only
7.1 – 9.0	+4	Week 1 only
9.1 – 10.0	+5	Week 1 only

→ From week 2 onwards the bump is fully removed — no carry-over.

Popularity Decay — Faster Post-Contract Drop

- Previous decay was far too slow (15%% → 5%% taking 10 weeks). Post-contract popularity now falls **5–7%% per week**, with the exact value randomised each week for realism.
- The randomised range means no two contract expirations feel identical.

Stream Generation — Dynamic Range (No More Static Output)

- The stream formula now produces a **computed base value** which is then spread across a dynamic range — the same inputs will never produce the exact same number twice.
- The spread range scales with the computed stream value — larger songs have wider variance:

Computed Streams	Output Range (approx.)	Variance Width
5,000	3,000 – 6,000	~60%% below / 20%% above
150,000	110,000 – 160,000	~27%% below / 7%% above
1,000,000	Proportionally wider	Range scales with base

Song Stream Decay — 4-Week Grace, 12-Week Floor

- Songs do not decay for the **first 4 weeks** after release — they draw from the full dynamic range.
- From week 5 onward, a decay variable activates and gradually reduces streams until they hit **5%% of the original computed base** by week 12.
- After week 12, the song settles into a permanent low range based on 5%% of its computed base — also randomised within a tight band.

Week Range	Stream Behaviour	Example (base: 5,000)
Weeks 1–4	Full dynamic range, no decay	3,000 – 6,000 / week
Weeks 5–12	Decay variable active, declining	Gradually falling toward 250
Week 12+	Settled at 5%% of base, with range	150 – 280 / week (permanently)

→ Further decay stages (e.g. after 6 months) are planned for a future update.

Popularity-Linked Decay Floor — Old Songs Still Benefit

- The 5%% decay floor is **not fixed to launch-week numbers** — it is recalculated each week against the current popularity-adjusted stream formula.
- This means a song released 12 weeks ago on low popularity can still generate strong numbers if the artist has grown significantly since release.

Scenario	Launch Week	Week 12 Popularity	Week 12 Streams
Low → High popularity arc	5%% pop → 5,000 base streams	80%% popularity	5%% of 50,000 = 2,500/wk
Flat popularity	5%% pop → 5,000 base streams	5%% popularity	5%% of 5,000 = 250/wk

→ An artist blowing up at week 12 breathes new life into their back catalogue — old songs get rediscovered as new fans explore their discography. Realistic and rewarding.

v1.3.
1

Introducing the Artist Ecosystem

13 April · 11:21 PM

Overview

- A pool of **50 pre-built artists** now exists in the world alongside the player. They release music, build popularity, react to the player, and interact with each other independently every week.
- Every ecosystem artist simulates alongside the player — the world keeps moving whether or not the player does anything.

Per-Artist Attributes

Attribute	Type	Description
name	string	Artist's display name
genre_proficiencies	dict {genre: int}	Skill level per genre (0–100), drives release quality
theme_proficiency	dict {theme: int}	Strength per theme — affects collaboration quality bonus
skills	dict (4 keys)	lyrics, vocals, production, mix/master — mirrors player skills
popularity	float 0–100	Current popularity in the ecosystem
release_tendency	float	How frequently this artist drops music (prob per week)
quality_consistency	float 0–1	How close to their ceiling they perform per release
friendliness	int 0–100	Governs personality type (friendly / neutral / rude) and ease of building relationship with player
controversy_meter	float 0–100	Higher = more likely to cause drama, appear in news, take jabs at the player or rivals

brutality	int 0–100	Response to disrespect: 0 = lets it go, 50 = disses back, 100 = escalates aggressively
first_release_week	int	The in-game week on which this artist drops their debut — staggers ecosystem so not everyone drops week 1
feature_cost	float (\$)	Price the player pays to feature this artist on a track

quality_consistency vs release_tendency — Now Separate Variables

- **release_tendency**: how often an artist drops music — a high value means they drop frequently, a low value means they go quiet for stretches.
- **quality_consistency**: how close to their ceiling they perform when they do release — an artist can drop rarely but always go 9/10, or drop every week but fluctuate wildly.
- These being separate creates realistic archetypes: the prolific-but-inconsistent artist, the slow-burning perfectionist, the hype machine who drops constantly but mid.

Theme Proficiency & Collaboration Bonus

- Each ecosystem artist has proficiency values per theme (heartbreak, street life, protest, etc.).
- When the player features an ecosystem artist on a track, the collaboration adds a quality bonus proportional to how well the featured artist's theme proficiency aligns with the song's theme.
- Example: Kendrick has high street life proficiency — featuring him on a street life track boosts quality more than featuring him on a party track.

Weekly Ecosystem Output — Release Feed

- After every week simulation, the game prints a release feed showing how many ecosystem artists dropped music that week, followed by individual release lines.
- Release line format:
→ **[Artist Name] releases [album / EP / single / mixtape] "[Generated Title]" — [Genre] / [Theme] — [Score]/10**
- The release title is fetched from the name generation file.
- The score is generated from the artist's skills, genre proficiency, and quality_consistency — integer only for ecosystem artists.

Staggered Debut System

- Each artist has a **first_release_week** variable — an in-game week number before which they will not release anything.
- This is randomised across the 50 artists at game start, spread across roughly the first 12–16 weeks, so the ecosystem builds organically rather than exploding all at once on week 1.
- After their debut, each artist then releases according to their release_tendency probability each week.
- The simulation logic mirrors the player's week loop — ecosystem artists tick forward on every simulate_week call, making the world feel synchronised.

Personality Types (derived from friendliness)

friendliness range	Personality Type	Behaviour
70–100	Friendly	Easy to build relationship with, less likely to jab, may reach out first
35–69	Neutral	Default state, reacts to what the player does, no strong lean

0–34	Rude	Hard to win over, higher base jab probability, low controversy threshold
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→ Reputation system (planned next) will expand artist interactions — *controversy_meter* and *brutality* are defined now but will fully activate once reputation is live.

Release Cooldown Periods — Format-Specific, Artist-Specific

- After releasing any project, an artist enters a cooldown period for that **format only**. Singles, EPs, mixtapes, and albums each have separate cooldown timers.
- Once the cooldown expires, the artist resumes releasing according to their *release_tendency*. Cooldowns are based on real release gaps derived from each artist's actual discography.

Artist	Single	EP / Mixtape	Album	Real-world basis
Frank Ocean	26 wks	80 wks	210 wks	Channel Orange (Jul '12) → Blonde (Aug '16) = ~212 wks. Singles sporadic since 2017.
Tyler, The Creator	12 wks	60 wks	120 wks	Flower Boy ('17) → Igor ('19) = ~104 wks; Igor → Call Me If You Get Lost ('21) = ~104 wks. Rare singles.
Kanye West	6 wks	30 wks	100 wks	MBDTF ('10) → Yeezus ('13) = ~130 wks; TLOP ('16) → ye ('18) = ~120 wks. Erratic singles.
Drake	2 wks	20 wks	52 wks	Albums ~every 12–18 months. Singles/loosies near-constantly. Views ('16) → Scorpion ('18) = ~110 wks but typically shorter.
J. Cole	10 wks	52 wks	96 wks	Born Sinner ('13) → FHD ('14) = ~78 wks; FHD → 4YEO ('16) = ~104 wks; KOD ('18) → The Off-Season ('21) = ~156 wks avg ~100.
Kendrick Lamar	8 wks	50 wks	130 wks	GKMC ('12)→TPAB ('15)=~130 wks; TPAB→DAMN ('17)=~104 wks; DAMN→Mr.Morale('22)=~260 wks; avg ~130.
Travis Scott	4 wks	35 wks	110 wks	Rodeo ('15)→Birds('16)=~52 wks; Birds→Astroworld('18)=~104 wks; Astroworld→Utopia('23)=~260 wks; avg ~140, skew lower.
21 Savage	4 wks	30 wks	70 wks	Issa ('17)→I Am>('18)=~52 wks; Savage Mode II ('20)→Her Loss('22, collab)=~110 wks. Active single dropper.
Future	2 wks	16 wks	52 wks	One of the most prolific in the ecosystem — drops albums near-annually, mixtapes/loosies constantly.
Lil Uzi Vert	3 wks	24 wks	80 wks	Luv is Rage ('15)→LiR 2('17)=~104 wks; Eternal Atake delayed significantly (~130 wks). Active single presence.
Playboi Carti	8 wks	50 wks	150 wks	Die Lit ('18) → Whole Lotta Red ('20) = ~130 wks; WLR → Music ('25) = ~220 wks. Notoriously slow dropper.
SZA	8 wks	50 wks	130 wks	CTRL ('17) → SOS (Dec '22) = ~286 wks. Very slow album cadence, sporadic singles.

Doja Cat	4 wks	30 wks	65 wks	Hot Pink ('19)→Planet Her('21)=~90 wks; Planet Her→Scarlet('23)=~115 wks. Active single presence.
Nicki Minaj	3 wks	30 wks	90 wks	Albums every ~2 years historically; Pink Friday 2 arrived after ~4 yr gap. Constant singles.
A\$AP Rocky	8 wks	40 wks	140 wks	ALLA ('15) → Testing ('18) = ~156 wks; Testing → Don't Be Dumb ('24) = ~320 wks. Infrequent dropper.
Metro Boomin	3 wks	20 wks	80 wks	Not All Heroes Wear Capes ('18)→Heroes & Villains('22)=~208 wks. Frequent producer features.
Juice WRLD	2 wks	16 wks	52 wks	Extremely prolific during his career — multiple loosies per month, albums ~annually.
Pop Smoke	2 wks	14 wks	40 wks	Very short career arc — Meet the Woo → Shoot for the Stars in ~26 wks. High frequency modelled.
Central Cee	3 wks	20 wks	65 wks	Wild West ('21) → 23 ('23) = ~100 wks. Active UK drill presence with consistent single drops.
Dave	8 wks	40 wks	110 wks	Psychodrama ('19)→We're All Alone in This Together('21)=~117 wks. Deliberate, infrequent dropper.

→ *Cooldown values are per-format and independent — an artist can be in album cooldown while still dropping singles freely.*

→ *All cooldowns are the **minimum** floor. release_tendency still governs whether a release actually fires once cooldown expires — so a low-tendency artist may wait several more weeks beyond cooldown before actually dropping.*